Bus Management System in C

- Pragathi Engineering College

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Introduction

- Console-based Bus Management System in C
- Stores bus details:
- - Bus Number
- - Driver Name
- - Route
- - Number of Seats
- Menu-driven program

Problem Statement

- Manual management is difficult
- Needs:
- - Efficient storage
- - Easy search & update
- - Reduced errors

Objectives

- Learn C programming
- Understand structures & arrays
- Implement CRUD
- Build menu-driven

CRUD refers to the four basic operations you can perform on data in your Bus Management System:

- C – Create → Adding new data

addBus() (adds a new bus record)

R – Read → Viewing data displayBus() (shows all bus details)

U – Update → Modifying existing data updateBus() (edits bus information)

D – Delete → Removing datadeleteBus() (removes a bus record)

Data Structure Used

- struct Bus {
- int busNo;
- char driver[50];
- char route[50];
- int seats;
- };
- Structure stores multiple data types

Array of Structures

- struct Bus buses[MAX];
- Array of structures stores multiple records efficiently
- struct Bus buses[2] = {{101,"Ramesh", "Hyderabad",50},
- {102, "Suresh", "Vijayawada", 45}};
- buses[0] → First bus
- buses[1] → Second bus
- Accessing DataCprintf("%d", buses[0].busNo);
- Use index + dot operator

Functions Used

- addBus()
- displayBus()
- searchBus()
- deleteBus()
- updateBus()
- Improves modularity

Control Statements

- while loop → menu
- switch-case → choices
- if → validation & search

Advantages & Limitations

- Advantages:
- Simple, user-friendly
- Limitations:
- No file storage
- No GUI
- Limited memory

Conclusion

- Demonstrates C concepts
- Uses structures, arrays, functions
- Implements menu-driven system